WORKED SOLUTION

STEP III 2008, Question 9

Mechanics • Friction and Variable Forces

STEP III 2008, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

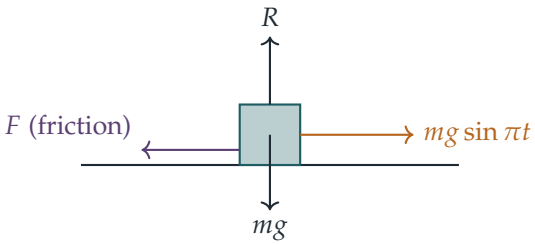
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The key idea

Friction has two regimes. While the particle is at rest, static friction matches the applied force exactly, up to its limit μmg, so the acceleration is zero. Once the particle slides, friction has the fixed size μmg and opposes the velocity. So the acceleration graph is made of flat zero pieces joined to shifted sine curves. "Comes to rest" means the velocity, which is the area under the acceleration graph since the start of the motion, has returned to zero.

Solution

We model the body as a particle on a rough horizontal plane. The normal reaction is $R = mg$, so friction has size at most μmg at rest and exactly μmg when sliding. Take the positive direction to be the direction of the force when $\sin \pi t > 0$, and let a and v be the acceleration and velocity.



Starting to move

Resolving vertically, $R = mg$. While the particle is at rest, horizontal equilibrium needs a friction force of size $mg|\sin \pi t|$, and this is possible as long as $mg|\sin \pi t| \leq \mu mg$. The particle first moves when the applied force first exceeds the limiting friction, so (for

$\mu < 1$)

$$\sin \pi T_0 = \mu, \qquad \text{that is,} \qquad T_0 = \frac{1}{\pi} \arcsin \mu, \quad 0 \le T_0 < \tfrac{1}{2}.$$

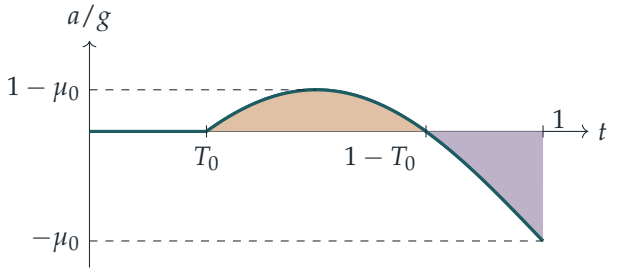
(If $\mu \ge 1$ the particle never moves.)

Part (i)

For $0 \le t \le T_0$ the particle is at rest, so $a = 0$. For $t > T_0$ it slides in the positive direction, friction $\mu_0 mg$ acts backwards, and Newton's second law gives

$$a = g(\sin \pi t - \mu_0) \qquad (\text{while } v > 0).$$

This starts at 0 when $t = T_0$, rises to $g(1 - \mu_0)$ at $t = \frac{1}{2}$, falls back to 0 at $t = 1 - T_0$ (by the symmetry $\sin \pi(1 - t) = \sin \pi t$), and is negative from then on, reaching $-\mu_0 g$ at $t = 1$.



The velocity is the area under this graph from T_0: it grows while the curve is above the axis (amber region) and shrinks after $1 - T_0$ (plum region). It stays positive until the two areas balance, and that is the moment the particle stops. So we need

$$v(1) = \int_{T_0}^{1} g(\sin \pi t - \mu_0)\, \mathrm{d}t = 0.$$

Now

$$\begin{aligned} \int_{T_0}^{1} (\sin \pi t - \mu_0)\, \mathrm{d}t &= \Big[-\frac{\cos \pi t}{\pi} \Big]_{T_0}^{1} - \mu_0 (1 - T_0) \\ &= \frac{1 + \cos \pi T_0}{\pi} - \mu_0 + \mu_0 T_0. \end{aligned}$$

Setting this to zero and multiplying by π:

$$1 + \cos \pi T_0 - \mu_0 \pi + \mu_0 \pi T_0 = 0.$$

Ask yourself Which sign does $\cos \pi T_0$ take when we write it in terms of μ_0?

Since $0 \le \pi T_0 < \frac{\pi}{2}$, the cosine is positive, so $\cos \pi T_0 = \sqrt{1 - \sin^2 \pi T_0} = (1 - \mu_0^2)^{1/2}$. Also $\pi T_0 = \arcsin \mu_0$. Substituting both,

$$1 + \left(1 - \mu_0^2\right)^{\frac{1}{2}} - \mu_0 \pi + \mu_0 \arcsin \mu_0 = 0,$$

as required. (Numerically $\mu_0 \approx 0.7246$ and $T_0 \approx 0.258$.)

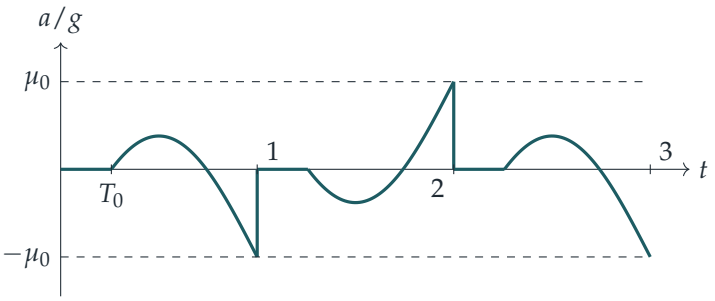
Part (ii)

At $t = 1$ the particle is at rest and the applied force is $mg \sin \pi = 0$. For $1 < t < 2$ the force points in the negative direction with size $mg|\sin \pi t|$. This is less than $\mu_0 mg$ until $|\sin \pi t| = \mu_0$, which first happens at $t = 1 + T_0$ (as $\sin \pi(1 + T_0) = -\sin \pi T_0$). So the particle stays at rest, with $a = 0$, for $1 \leq t \leq 1 + T_0$.

After that it slides in the negative direction, friction acts in the positive direction, and

$$a = g(\sin \pi t + \mu_0).$$

Since $\sin \pi(t + 1) = -\sin \pi t$, this is exactly the negative of the acceleration at time $t - 1$ in part (i). The motion on $[1, 2]$ is therefore the mirror image of the motion on $[0, 1]$: the velocity is the negative of the earlier one, so the particle comes to rest again at $t = 2$, having travelled back the same distance. It is then at rest with zero applied force, which is exactly the situation at $t = 0$, so on $[2, 3]$ the motion of $[0, 1]$ repeats.



The vertical segments at $t = 1$ and $t = 2$ mark jumps in the acceleration: at the instant the particle stops, kinetic friction (of size $\mu_0 mg$) is replaced by static friction, which only has to balance a zero applied force.

Motion for $\mu = \mu_0$. The particle is at rest until $t = T_0$, slides forwards and stops at $t = 1$, waits until $t = 1 + T_0$, slides back and stops at its starting point at $t = 2$, and so on. It shuttles back and forth over the same stretch of ground with period 2, pausing for time T_0 at each end. (Integrating once more gives the length of the stretch as $g[\mu_0/\pi^2 - (1 - T_0)(1 - \sqrt{1 - \mu_0^2})/(2\pi)] \approx 0.0367g$, though this is not asked for.)

Motion for $\mu = 0$. Now $T_0 = 0$ and $a = g \sin \pi t$ at all times, so

$$v = \frac{g}{\pi}(1 - \cos \pi t) \geq 0, \qquad x = \frac{g}{\pi}\left(t - \frac{\sin \pi t}{\pi}\right).$$

The velocity is never negative and vanishes only at $t = 0, 2, 4, \ldots$. So the particle never turns back: it moves forwards in surges, momentarily at rest at even times, and advances $2g/\pi$ in each period of 2, an average speed of g/π. Friction is what stops it at $t = 1$ in the rough case; without friction the backward half of the force only slows the particle down to zero velocity.

Checks

Solving the equation of part (i) by bisection gives $\mu_0 \approx 0.72461$, and so $T_0 \approx 0.25798$. A numerical simulation of the motion for $0 \le t \le 3$ with $g = 9.8$, switching between static and kinetic friction, shows the particle starting at $t \approx 0.258$, stopping at $t = 1.000$ after travelling 0.3598, starting back at $t \approx 1.258$ and stopping at $t = 2.000$ at the starting point, in agreement with $0.0367g$. For $\mu = 0$ the simulation matches $x = g(t - \sin \pi t/\pi)/\pi$. A quick sanity check on the equation: at $\mu_0 = 0$ the left side is $2 > 0$ and at $\mu_0 = 1$ it is $1 - \pi/2 < 0$, so a root in $(0,1)$ exists.

Common slips

- Taking $a = g(\sin \pi t - \mu)$ from $t = 0$: before T_0 the particle is at rest and friction is static, so $a = 0$.
- Keeping the friction in the same direction after the particle reverses in part (ii). Friction always opposes the velocity.
- Thinking the particle stops at $t = 1 - T_0$, where the acceleration (not the velocity) is zero.

Beyond STEP

Friction that switches between "stuck" and "sliding" makes the equation of motion piecewise and the acceleration discontinuous. This is the stick–slip behaviour behind squeaking doors and bowed violin strings, and a first example of a non-smooth dynamical system. Kleppner and Kolenkow, *An Introduction to Mechanics*, discusses friction models and forced motion at the right level.

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